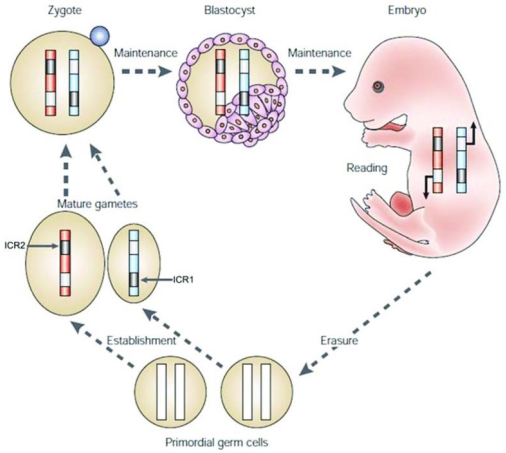
BME 105: Genetics in the Genomics Era

Lecture 15: Gene regulation: methylation and imprinting



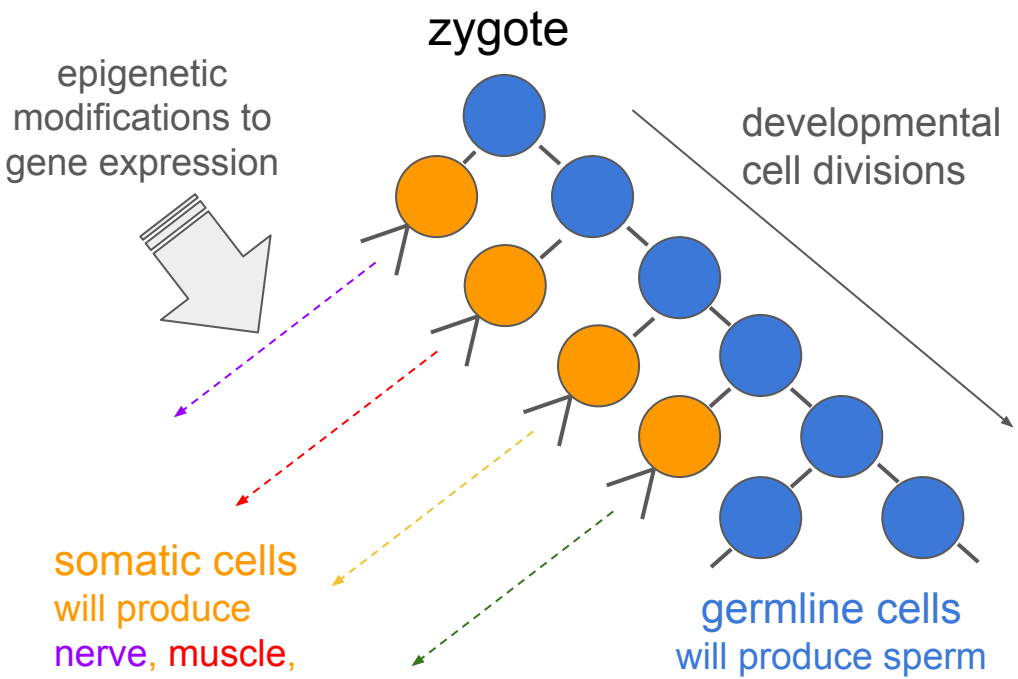
Instructor: Dr. Shelbi Russell,
Assistant Professor

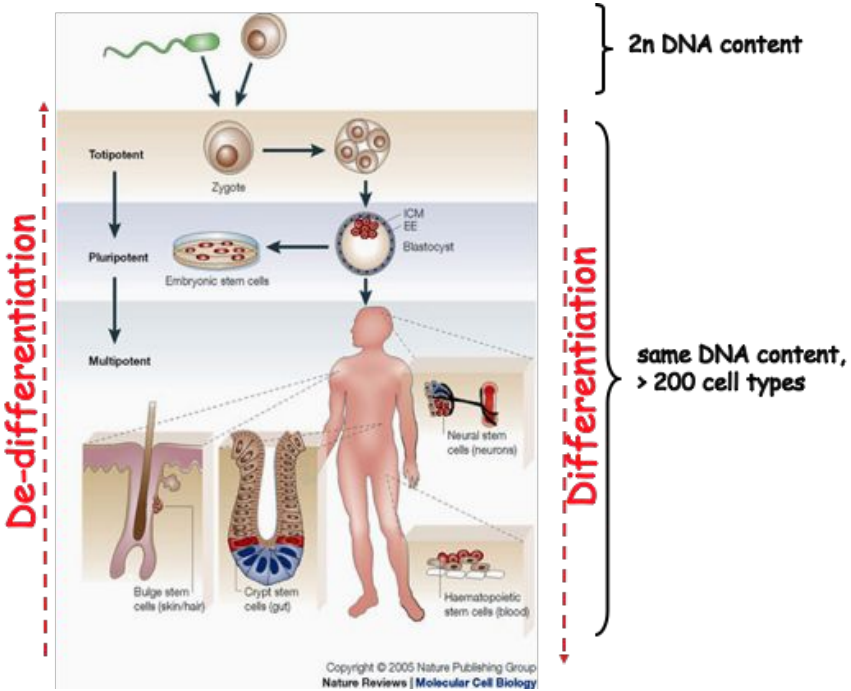
Course website on canvas.ucsc.edu

Lecture 15 Learning Objectives

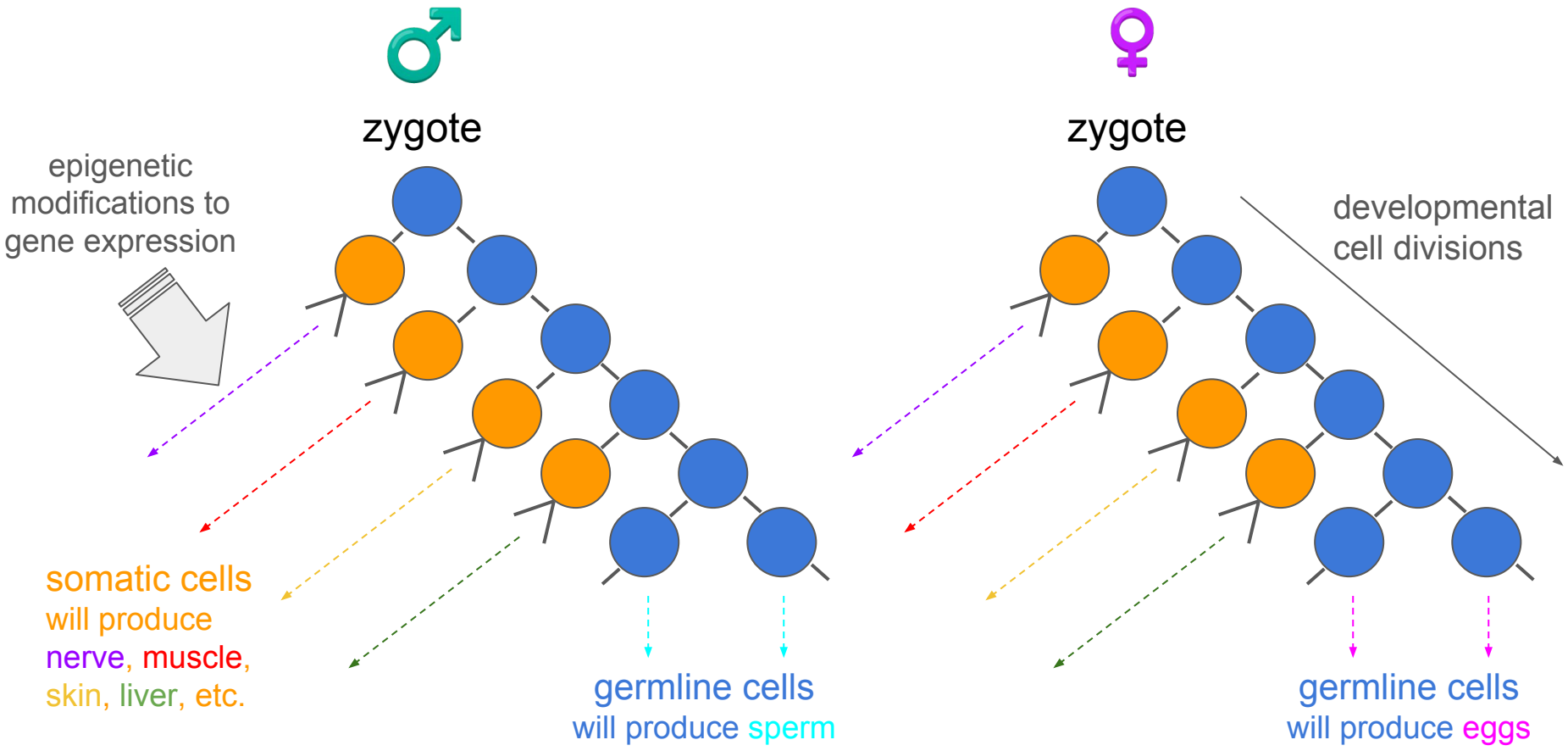
- Understand how DNA methylation regulates the genome:
  - By closing chromatin and blocking transcription
  - By blocking the transcription of ncRNAs that block transcription, thereby promoting transcription
- Understand how CpG islands recruit both methylation (at Cs) and activators (to the region) that repress methylation
- Understand the definition and implications of genomic imprinting
  - DNA methylation is epigenetically inherited through mitosis and is rewritten during meiosis
- Integrate pedigree analysis with epigenetic imprinting information to predict the probability of carrier and affected offspring
- Understand how the environment might influence the epigenome, and how you could test hypotheses
  - Learn one new sequencing approach: bisulfite sequencing - unmodified Cs are edited to Us, whereas methylated-Cs are not
  - Understand that long read data have more information in them than just the base pairs: Nanopore and PacBio can detect methylation modifications

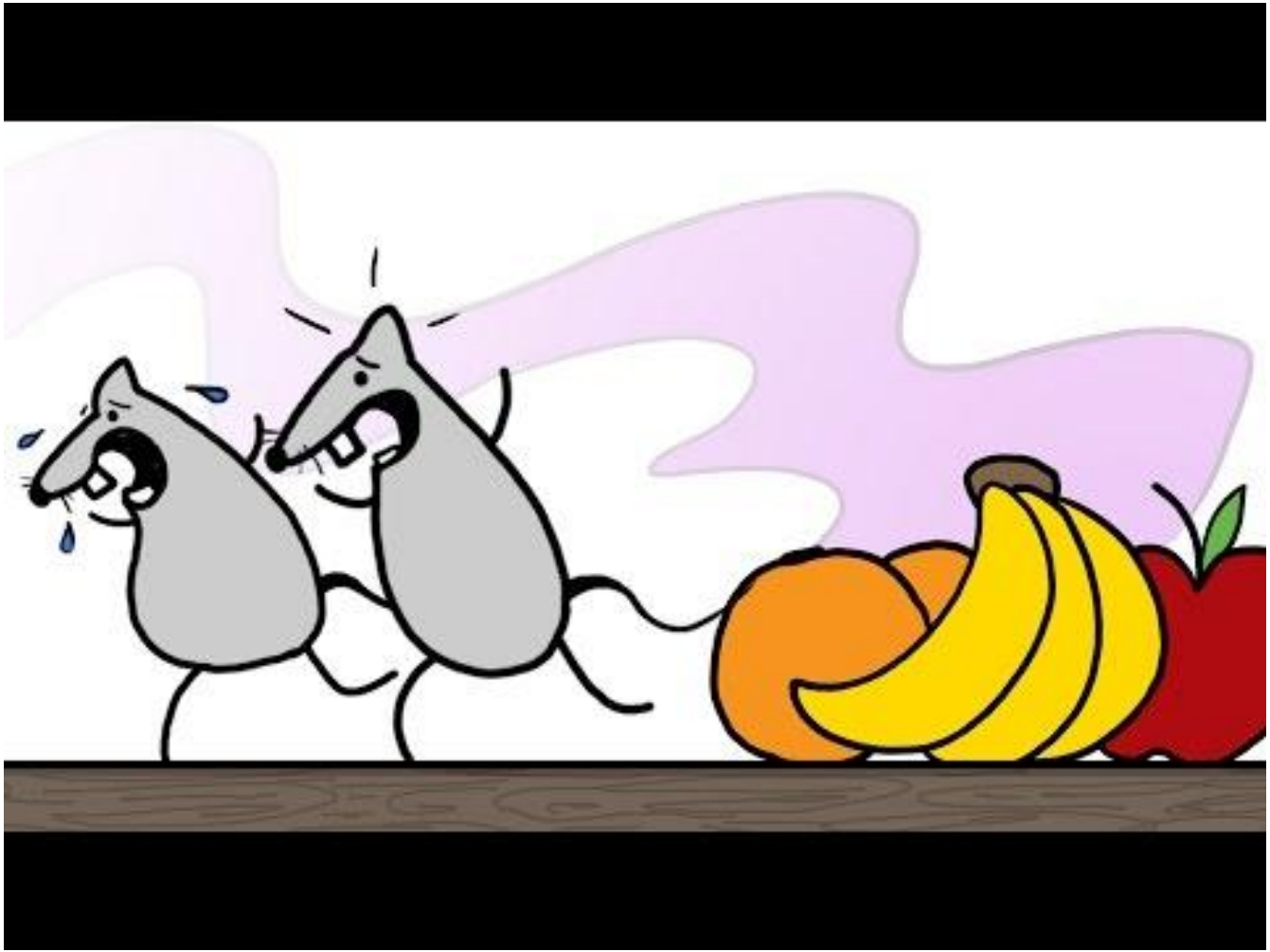
Epigenetic regulation via the non-coding genome during development



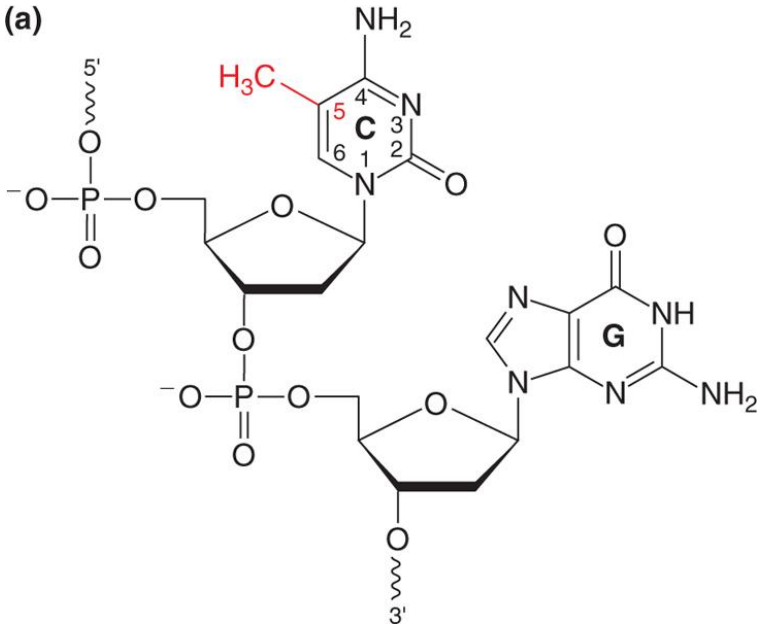


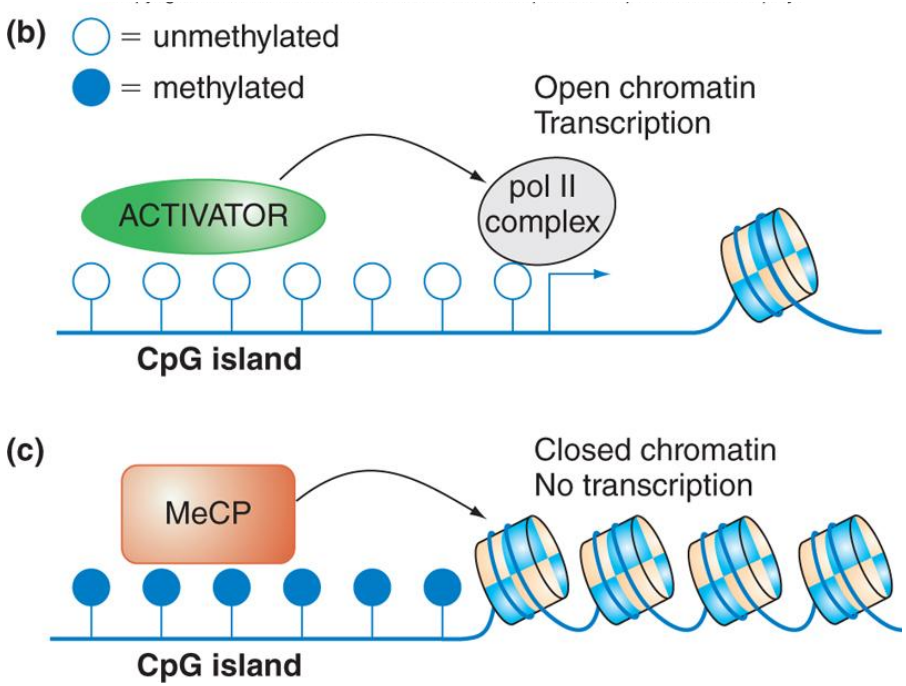
Epigenetic regulation via the non-coding genome during development





Control of expression through DNA methylation





*DNA methylation is often, but not always, associated with repressed transcription

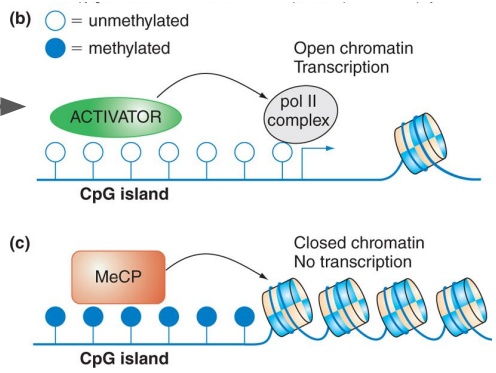
- CpG island - stretch of repeated CpG sequences
- MeCP - *methyl-CpG-binding proteins* (*MeCPs*)
  - Bind to methylated CpGs and close chromatin (*i.e.,* a methylation feedback loop)

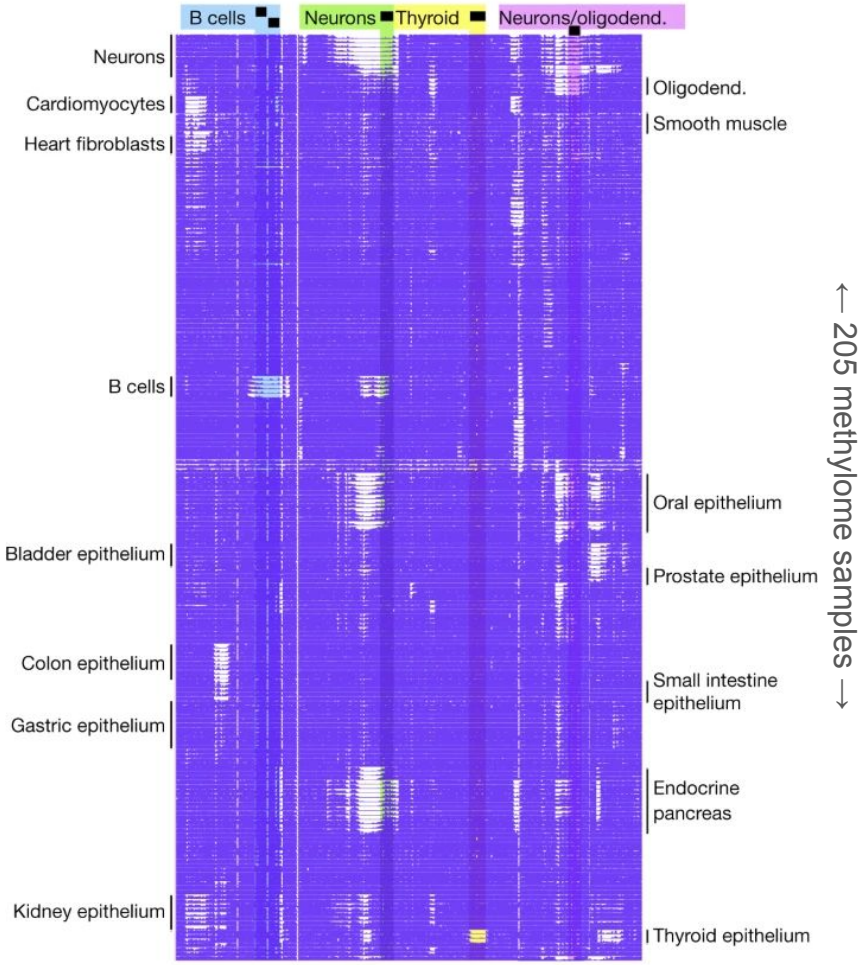
DNA methylation is important for housekeeping gene expression in vertebrates, esp. humans

~70% of CpGs are methylated

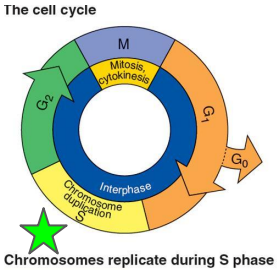
CpG islands can also block methylation

- By binding tissue-specific activators that block C-methylation
- 100-1000s base pairs long

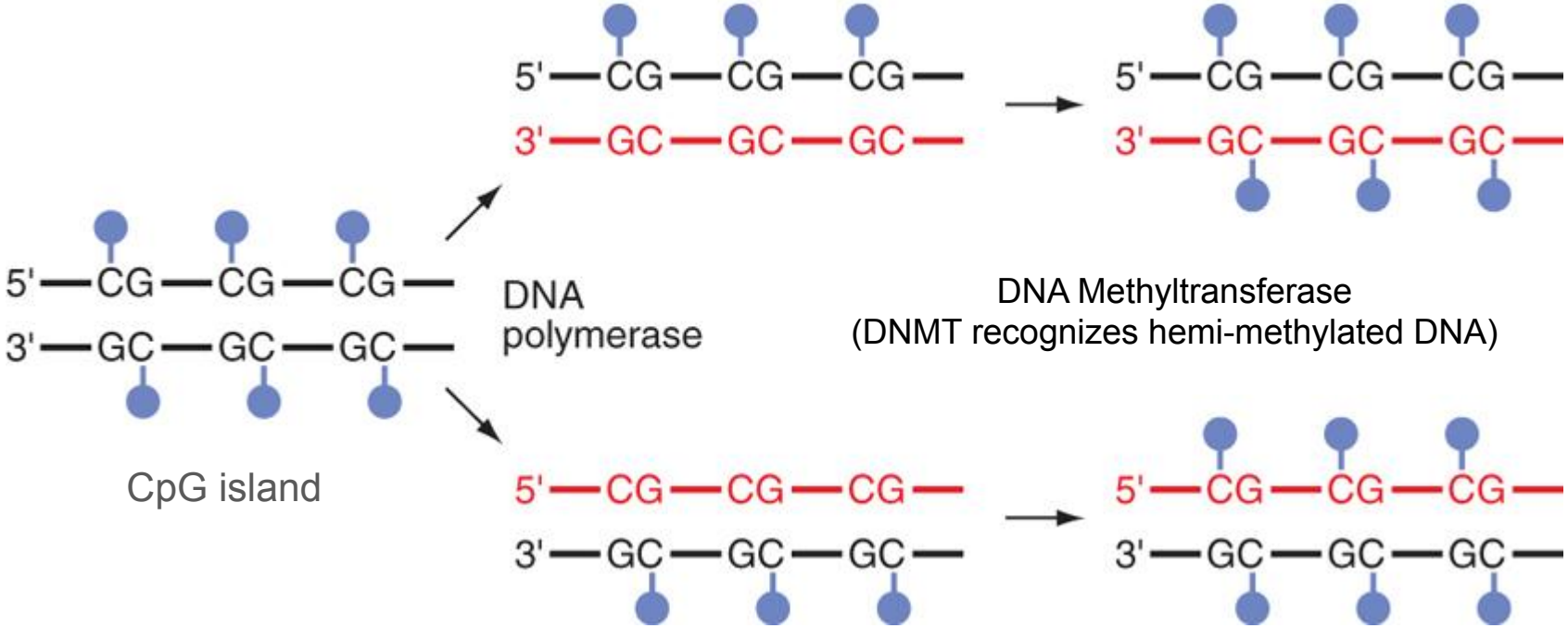




Loyfer et al. 2023



Cytosine methylation is perpetuated during DNA replication



Genomic imprinting: differential expression of parental alleles

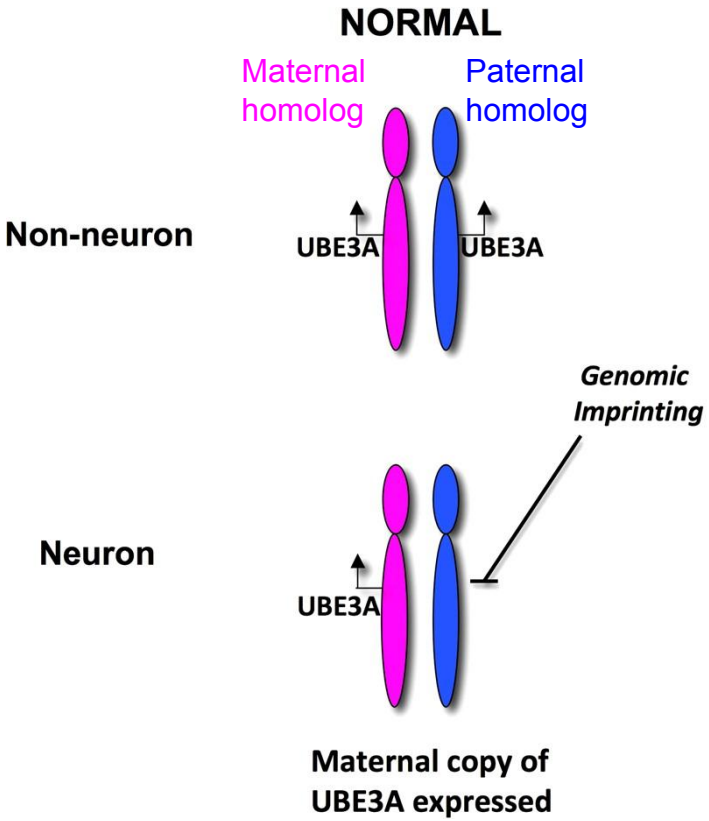
Imprinting - genes are expressed in a parent-inherited manner.

The imprinted copy is silenced through **sex-specific methylation** of imprinting control regions (ICRs)

Imprinted ~ silenced

Genes may be <u>maternally or paternally</u> imprinted

Only about 100 genes in the human genome exhibit imprinting



Chamberlain and Lalande 2010

Genomic imprinting: differential expression of parental alleles

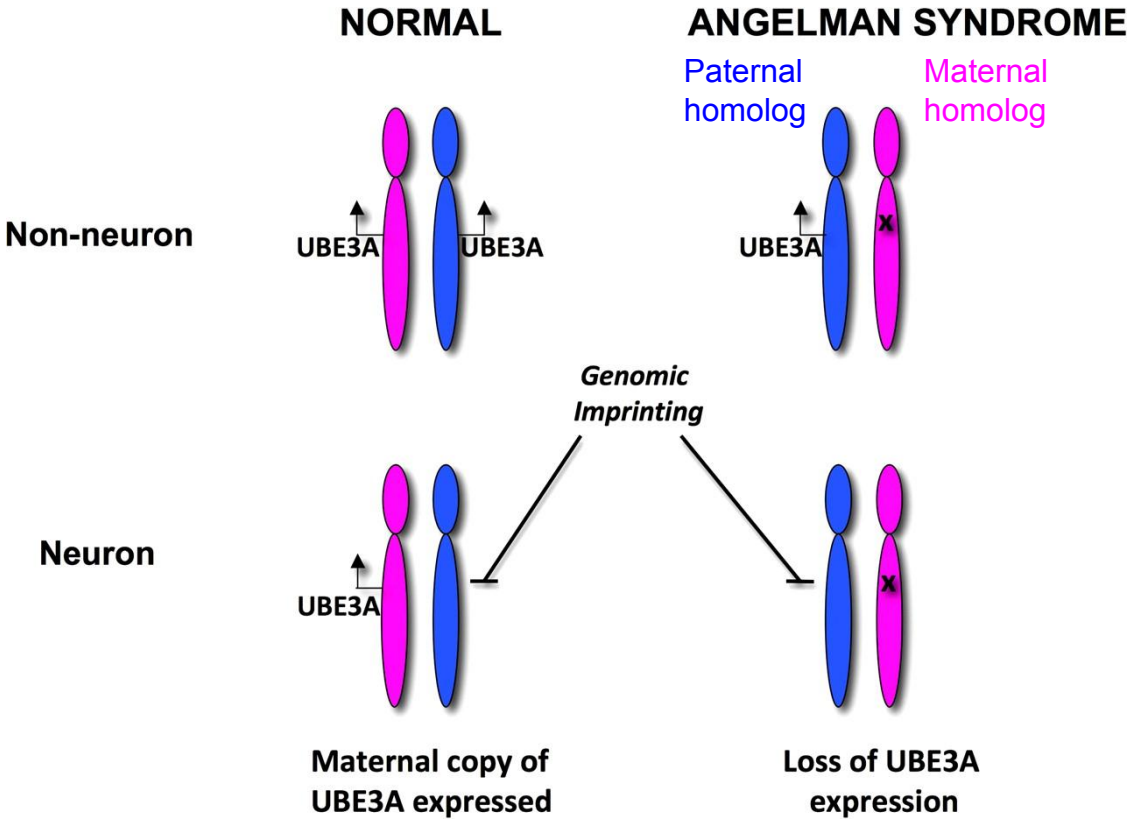
Imprinting - genes are expressed in a parent-inherited manner.

The imprinted copy is silenced through **sex-specific methylation** of imprinting control regions (ICRs)

Imprinted ~ silenced

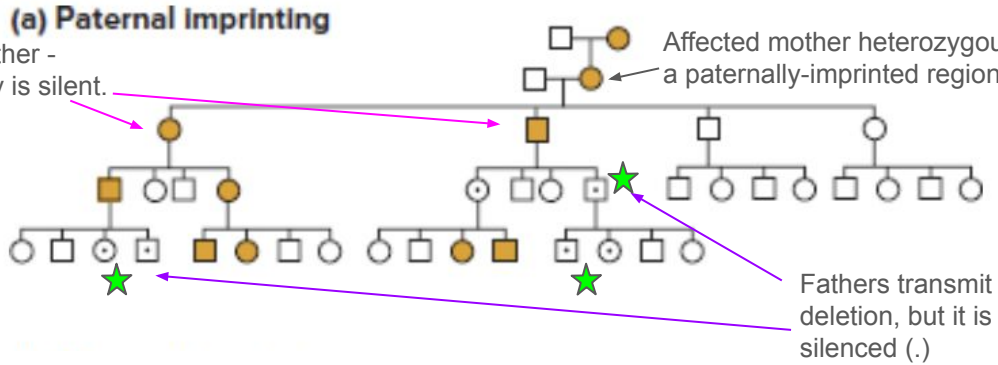
Angelman Syndrome example:

- The gene is <u>paternally imprinted</u>
  ⇒ paternal allele silenced
- Loss of UBE3A expression:
  - Maternal allele should be expressed, but it is mutated
  - No <u>expression is possible from a deletion</u>
  - The individual is affected, despite paternal allele being intact because it is silenced



Chamberlain and Lalande 2010

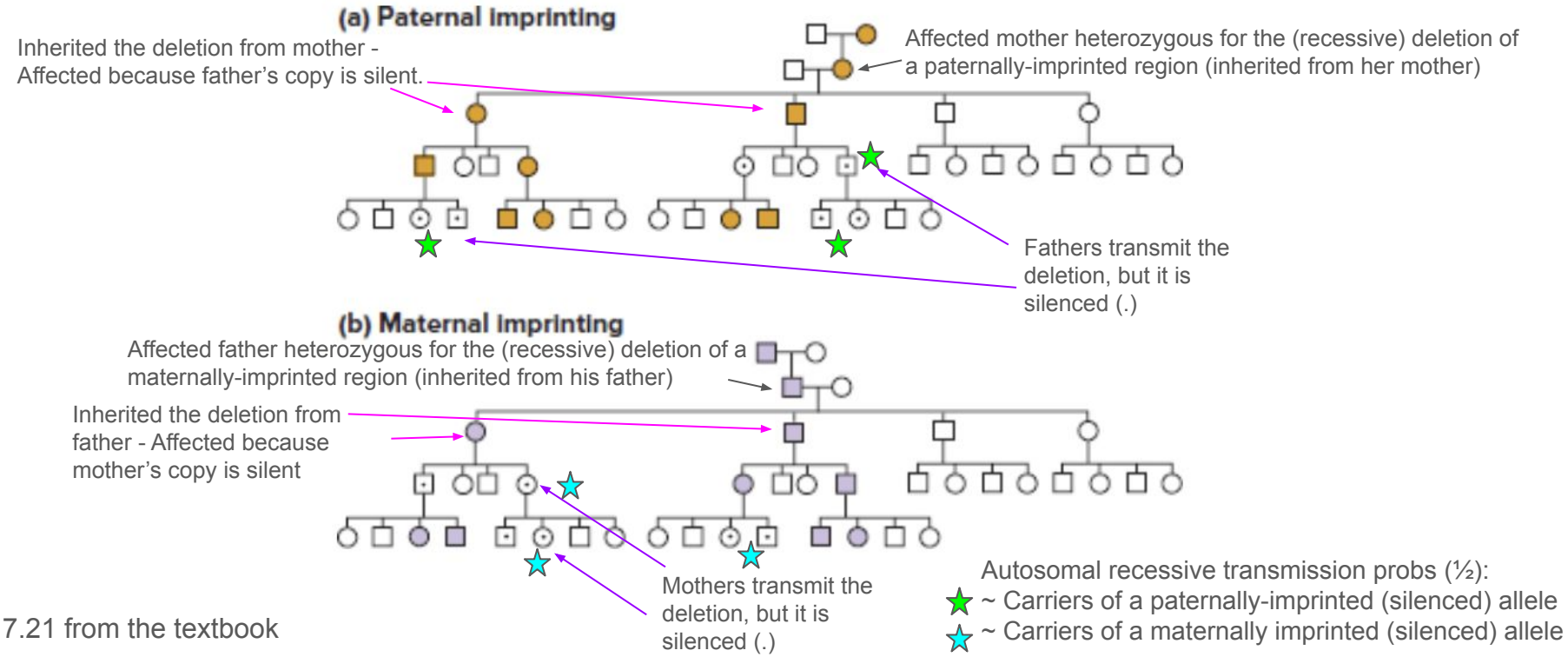
Inheritance of autosomal deletion phenotypes



Autosomal recessive transmission probs (½):
~ Carriers of a paternally-imprinted (silenced) allele
~ Carriers of a maternally imprinted (silenced) allele

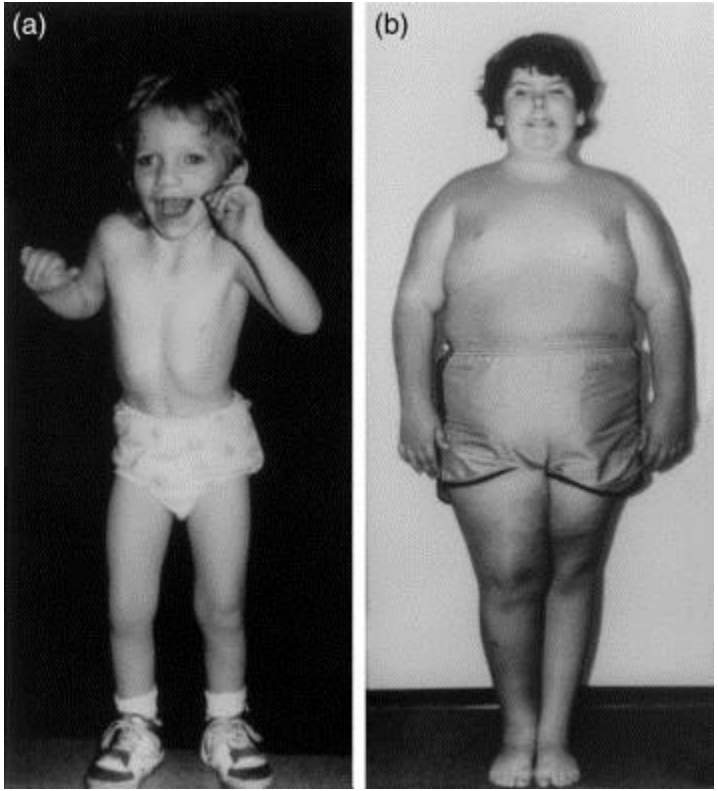
17.21 from the textbook

Inheritance of autosomal deletion phenotypes



17.21 from the textbook

Prader-Willis vs Angelman Syndrome



Clinical phenotype of imprinted disorders:

(a) Angelman syndrome. The characteristic happy disposition, laughter, widely spaced teeth and wide mouth, and stiff, upheld arms and broad stance are exhibited.

Genetics:

- Imprinted paternal chromosome.
- Deletion inherited from mother.

(b) Prader–Willi syndrome. This patient displays the typical central obesity, short stature, small hands and feet, and mild facial dysmorphism.

Genetics:

- Imprinted maternal chromosome.
- Deletion inherited from father.

Nicholls et al. Trends in Genetics, 1998

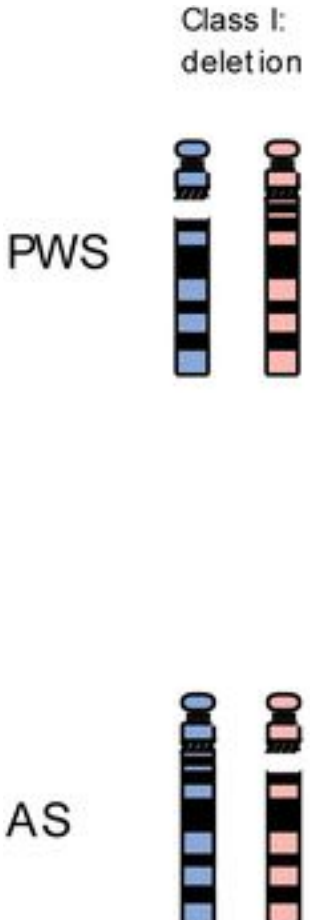
Prader-Willis Syndrome (PWS) vs Angelman Syndrome (AS)

The same region involved in Angelman Syndrome is also associated with Prader-Willis Syndrome.

- **Requires at least two genes in this region that are differentially imprinted**

PWS: Multiple paternally expressed, genes associated with the PWS phenotype are in this region. Region is imprinted in the maternal chromosome and deleted in the paternal chromosome.

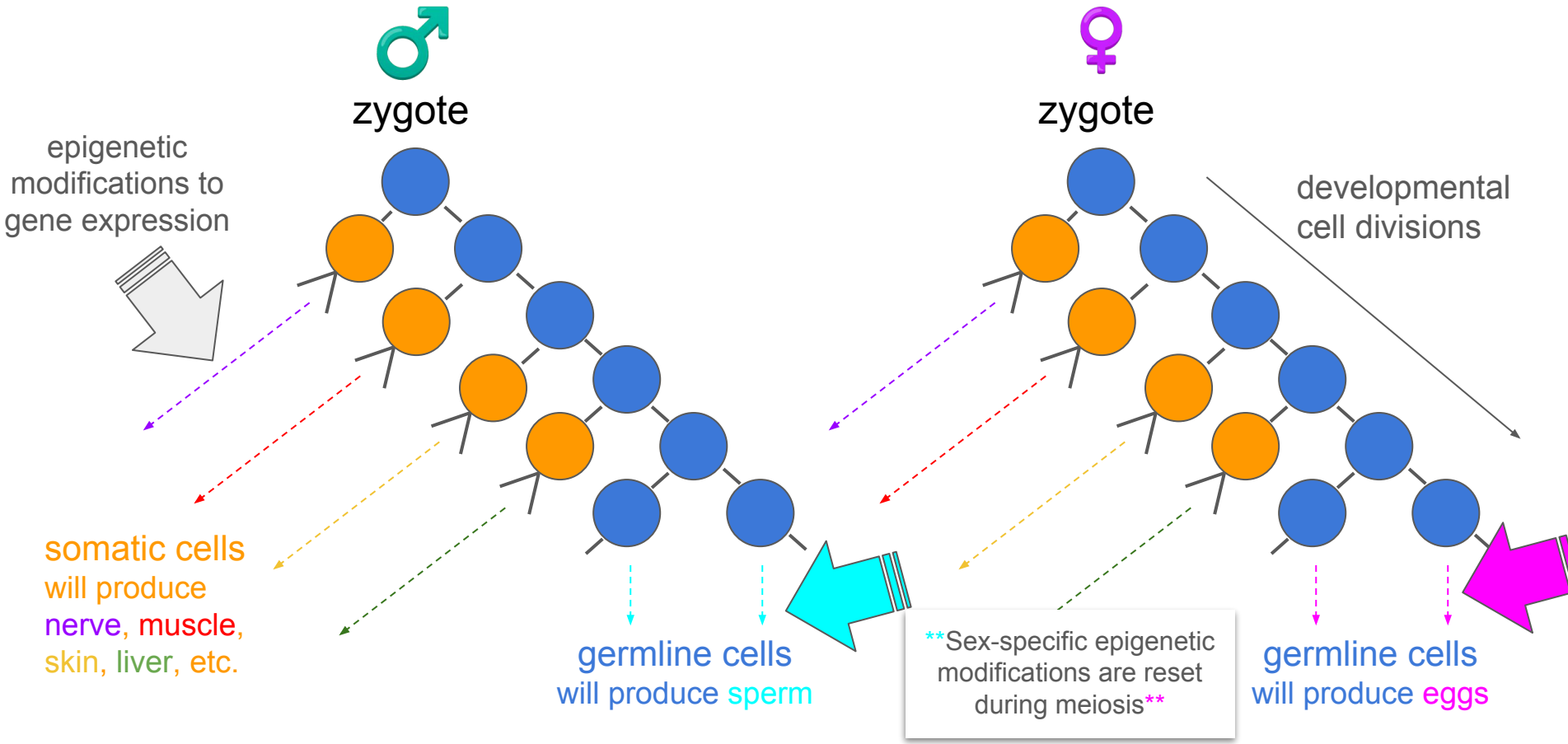
AS: A single maternally expressed gene associated with AS. Region is imprinted in the paternal chromosome and deleted/mutated in the maternal chromosome.



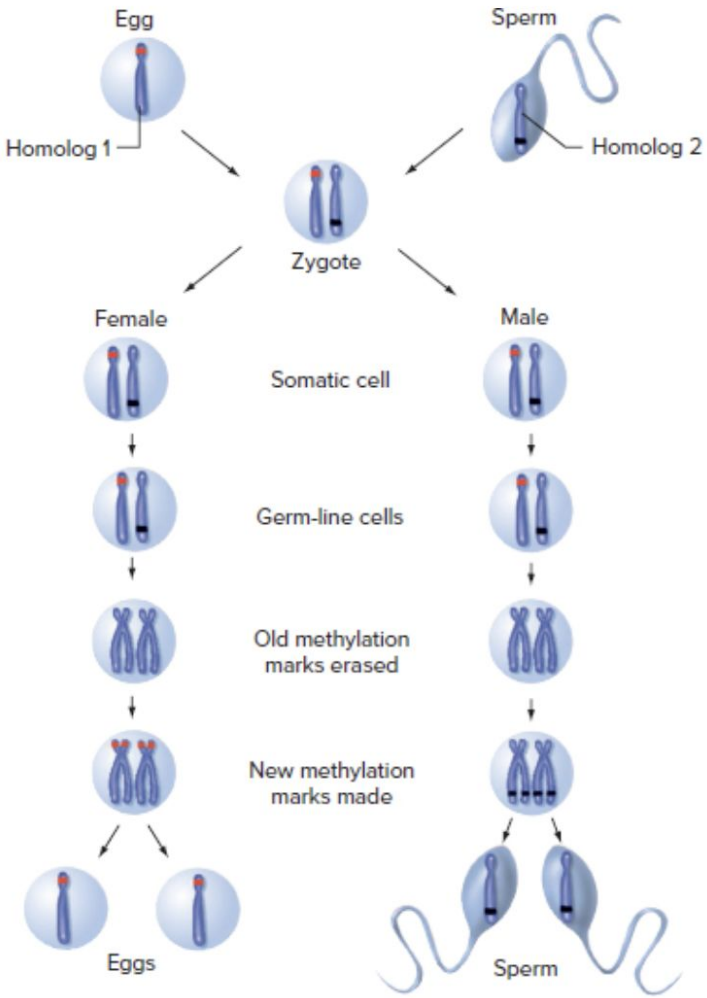
Blue - paternal chromosome

Pink - maternal chromosome

Epigenetic regulation via the non-coding genome during development



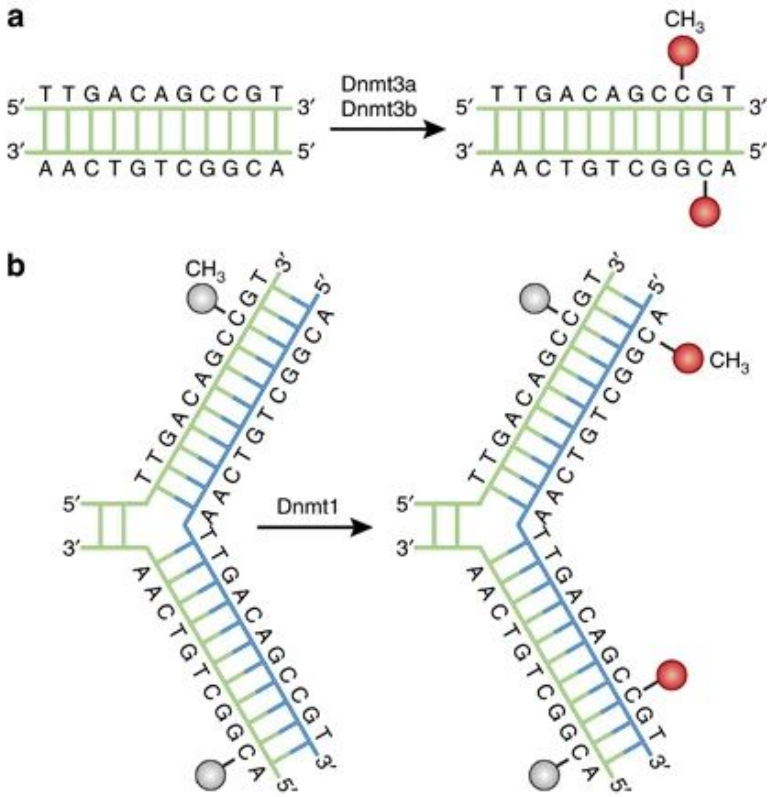
Genomic methylation marks are reset during meiosis



De novo vs maintenance methylation uses different Dnmts

De novo
methyl groups

Maintenance
methylation



Moore et al. 2013

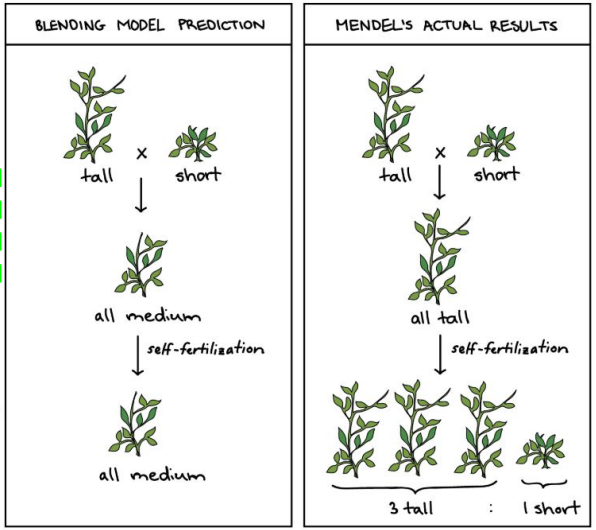
Imprinting is a big deal! For ~100 genes (in mammals), Mendel's investigations didn't tell the whole story:

From Lecture 1:

Mendel's investigations of the patterns of inheritance

Two (incorrect) assumptions in the mid-1800s:

- ~~One parent (the male) contributes more heritability~~
- ~~Parental traits become mixed forever by blended inheritance~~

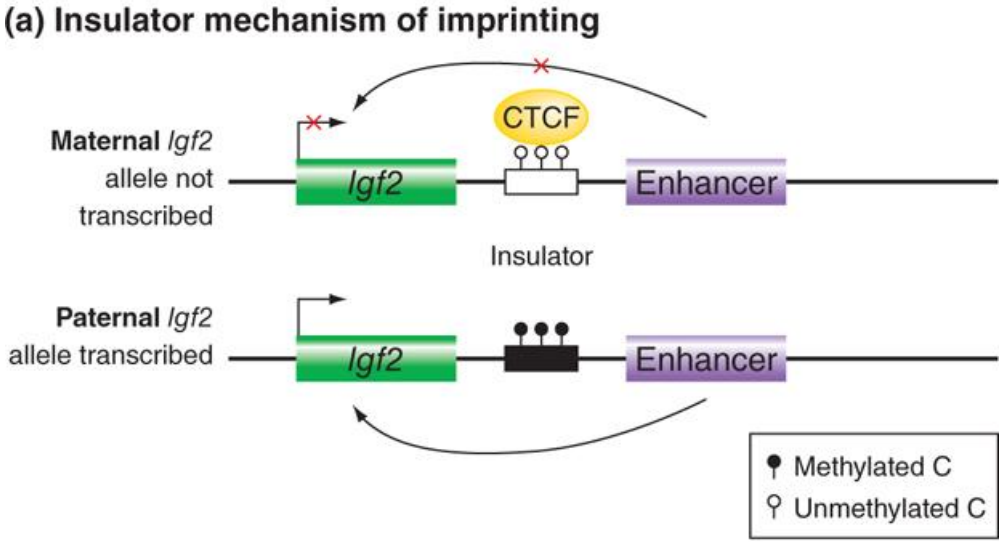


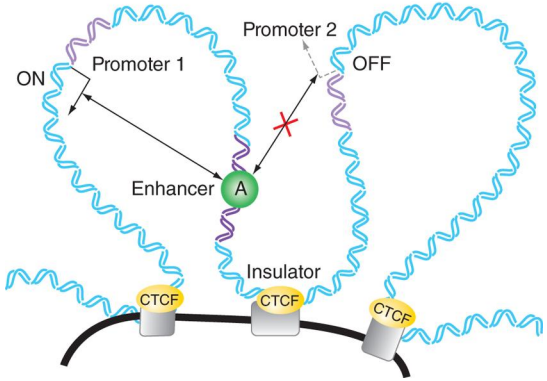
- Both parents contribute equally
  → one allele from each
  (except, see organellar inheritance)
- Traits remain intact
  → recessive alleles reappear in F2s

Two known genomic imprinting mechanisms

a) Methylation of insulator sequences prevents their binding to CTCF proteins, blocking loop formation

⇒ DNA methylation turns on expression (*Igf2's* enhancer can interact)





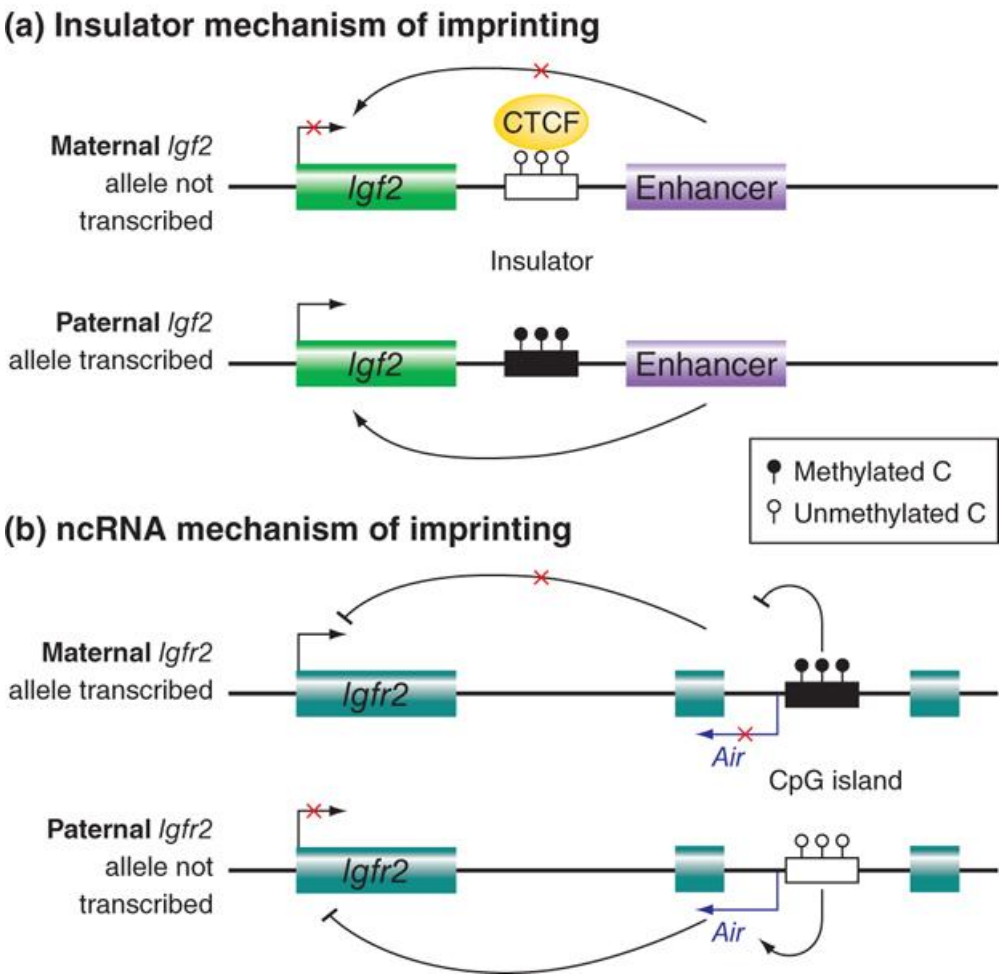
Two known genomic imprinting mechanisms

a) Methylation of insulator sequences prevents their binding to CTCF proteins, blocking loop formation

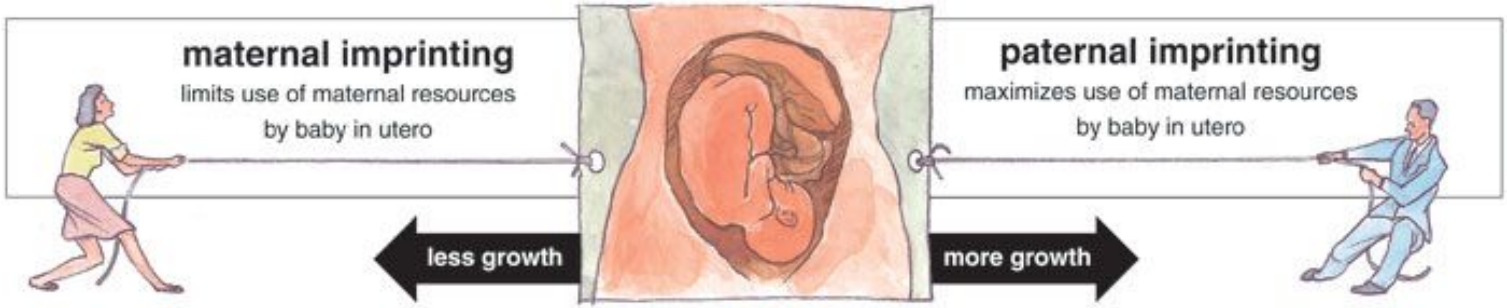
⇒ DNA methylation turns <u>on</u> expression (*Igf2's* enhancer can interact)

b) Non-coding RNA expressed in males blocks expression of the ORF that is its reverse complement

⇒ DNA methylation turns <u>off</u> ncRNA expression, which turns on *Igfr2*



Paternal conflict hypothesis of imprinting



⇒ Selection for spreading mom's resources among offspring

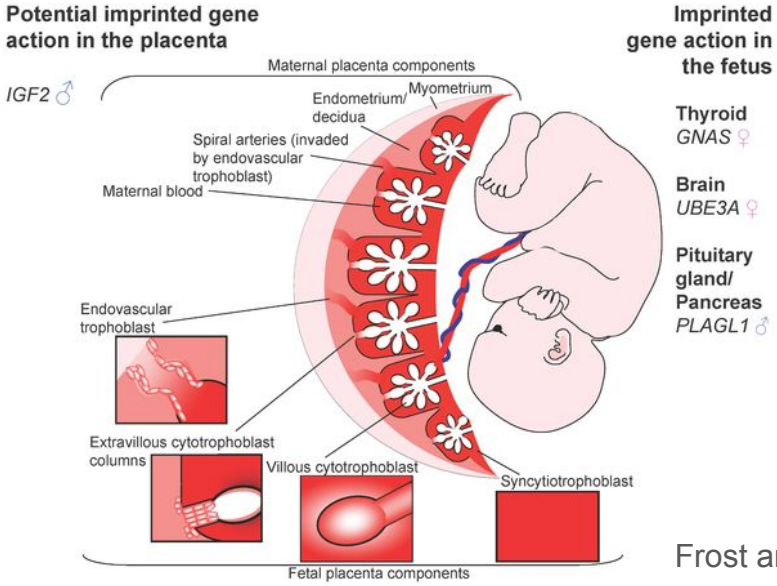
⇒ Selection for maximizing the resources the [father's] offspring extract from mom

Paternal conflict hypothesis of imprinting

Villous trophoblasts of the human placenta grow as a branched structure, maximising exchange with maternal blood.

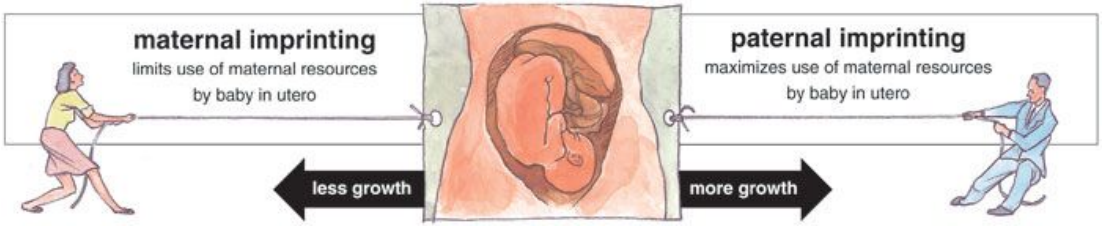
Extravillous trophoblast invade into the maternal endometrium, and some cells colonise maternal spiral arteries, expanding them to maximise blood flow.

Imprinted genes are important during fetal growth

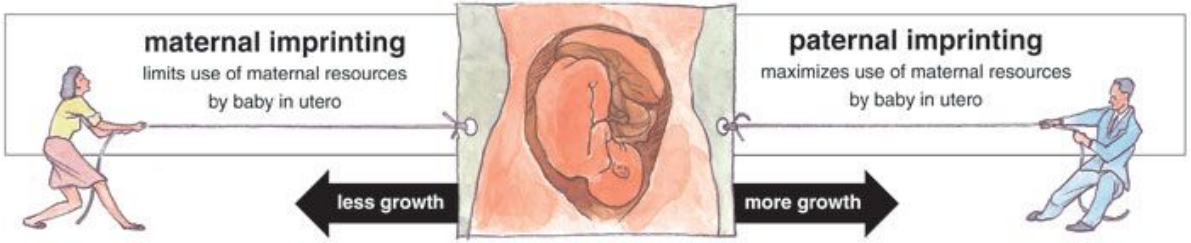


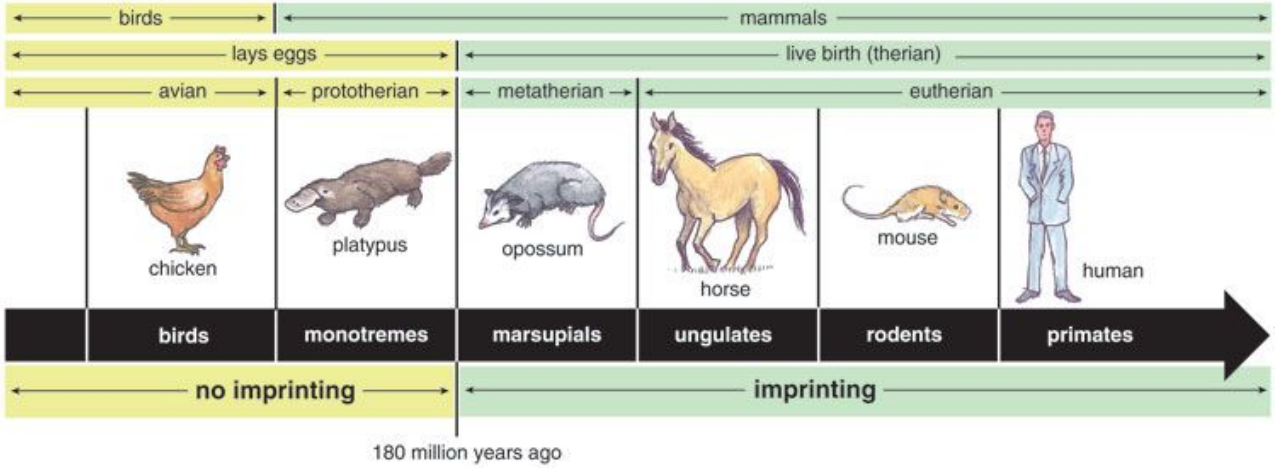
♂ = Paternally expressed
♀ = maternally expressed

Frost and Moore 2010



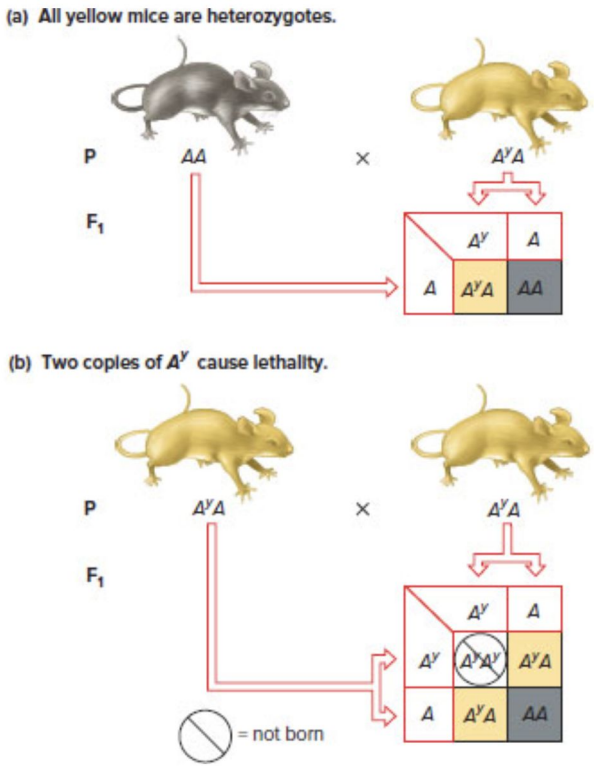
Paternal conflict hypothesis of imprinting, and its recent evolution in (live bearing) mammals

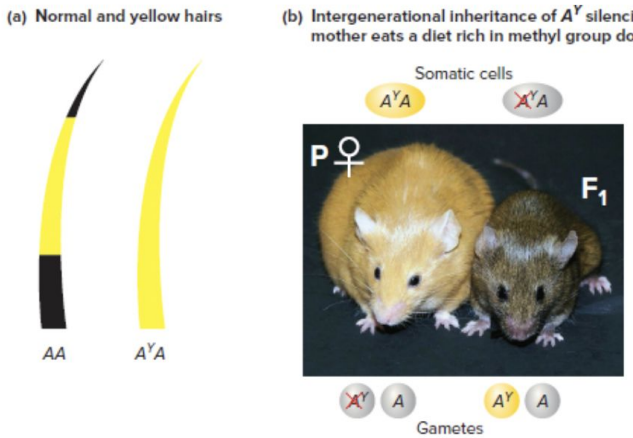




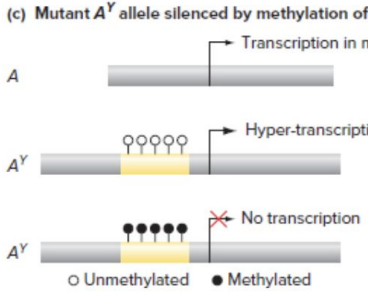
https://www.americanscientist.org/article/imprinted-and-more-equal

Environmental impacts: Intergenerational epigenetic inheritance





F0 mother fed garlic and beets, which are rich in CH3-donors



A^y is <u>environmentally</u> silenced
→ dominant allele is masked in F1s, when inherited from parent fed CH3 donors

Allele A^y induces the upregulation of pheomelanin in melanocytes

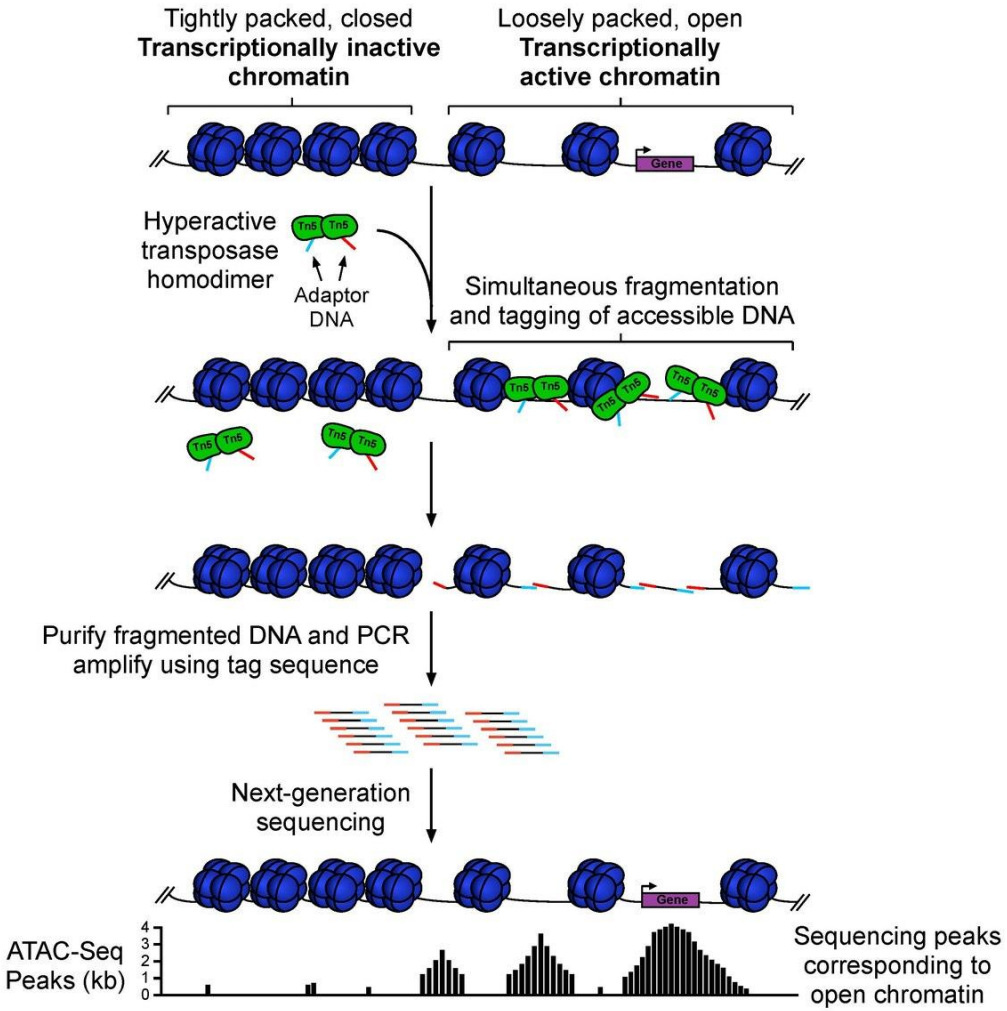
How do we identify regulatory regions?

Genomics project to identify epigenomic features

ATAC-Seq

Quick review from last lecture:

Detecting DNA accessibility

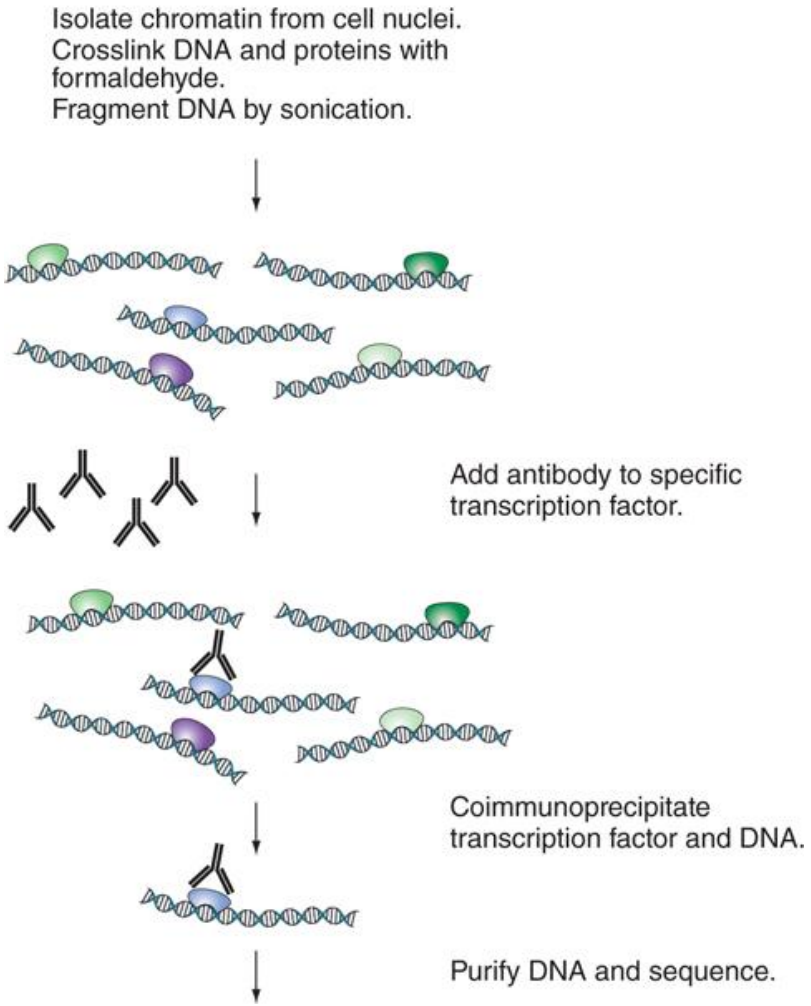


Wikipedia

ChIP-Seq

Quick review from last lecture:

Detecting binding of transcription factors



Chromatin contact sequencing: 3C, Hi-C, Micro-C, etc.

3C == Chromatin Conformation Capture

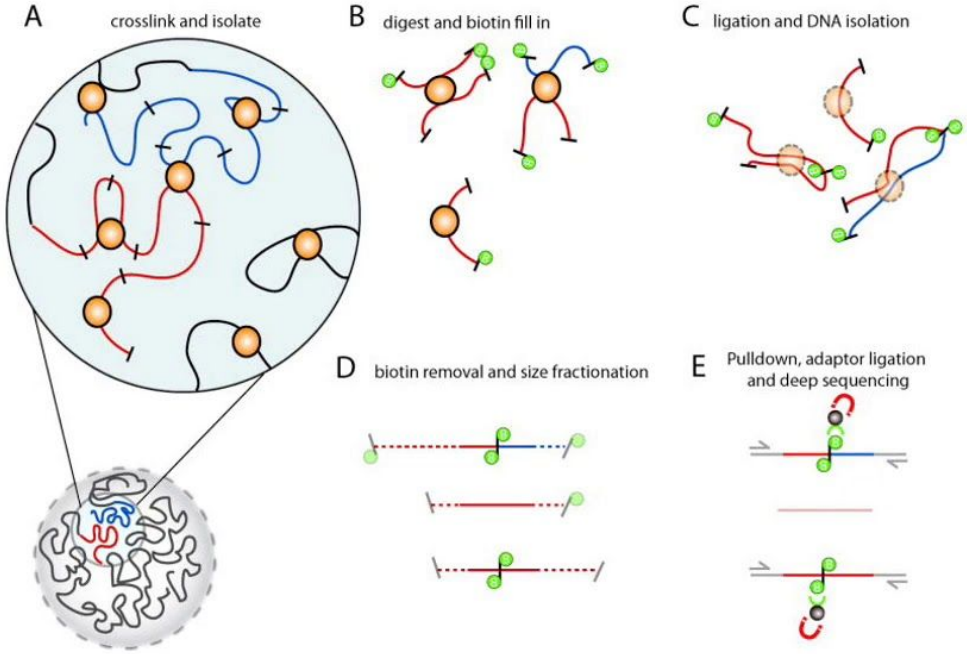
X-C are derivations of the original protocol

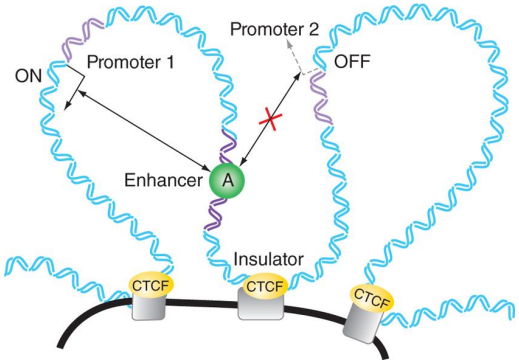
Quick review from last lecture:

Linking genes to regulatory regions

Hi-C: all-by-all interactions

- **Note: Biotin labels fragment ends without interfering with folding of bound proteins, DNA**





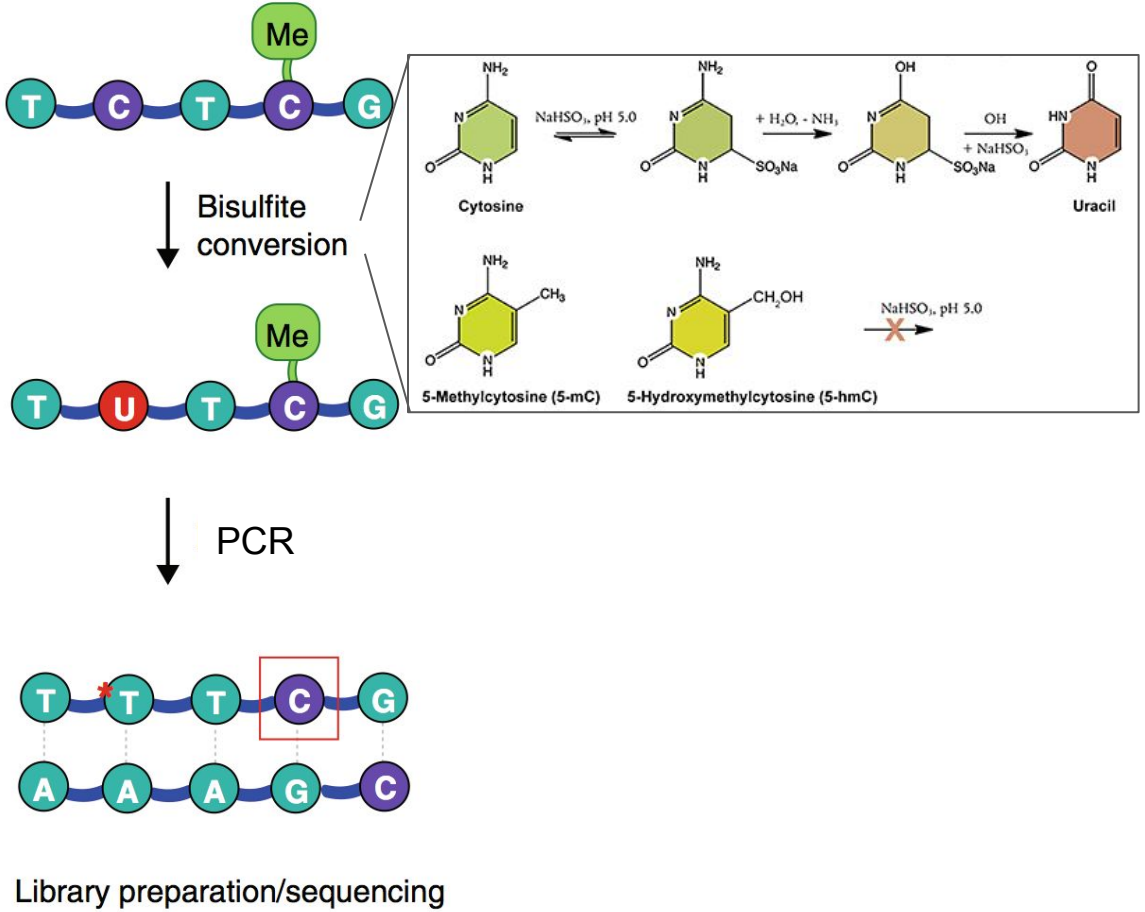
https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3874846/

Bisulfite sequencing

Methylated Cs are protected from bisulfite conversion.

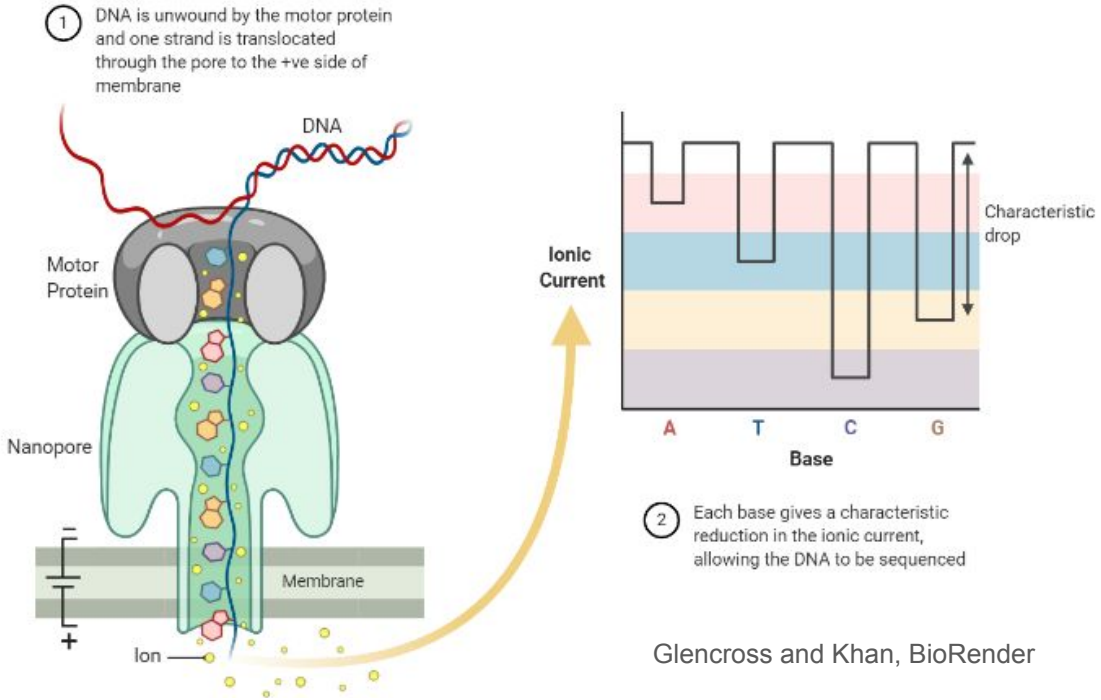
Unprotected Cs will be changed to U

In sequencing, unprotected Cs will have a sequence change from a C->T



Nanopore sequencing can detect DNA methylation natively

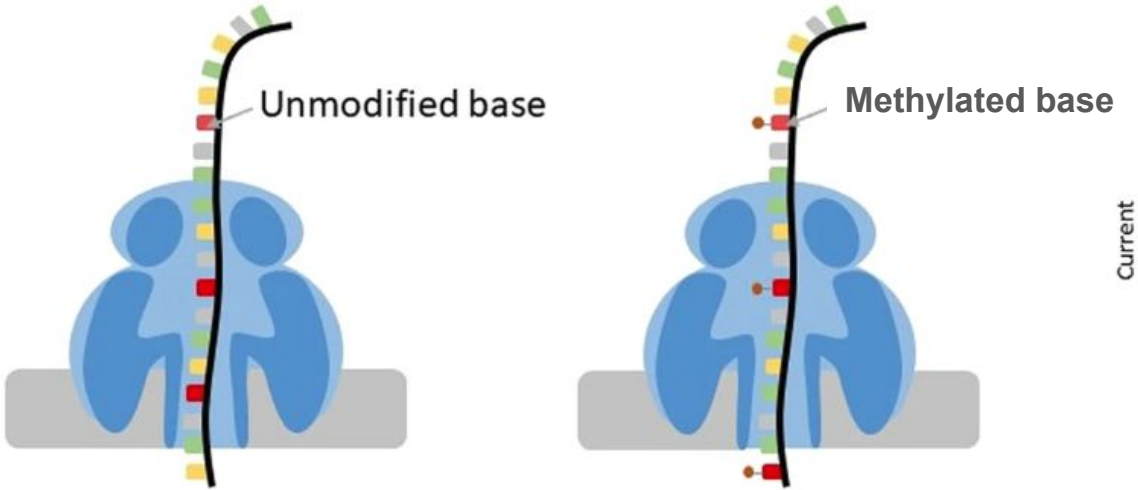
Review:

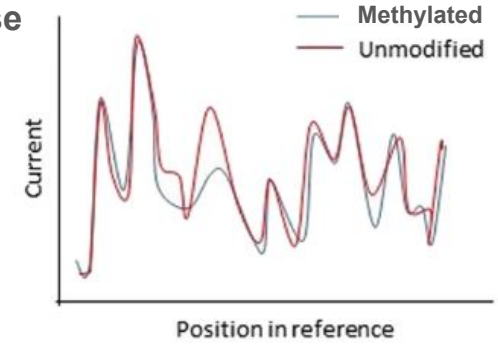


Glencross and Khan, BioRender

Nanopore sequencing can detect DNA methylation natively

Methylation is detected as a difference in current compared to unmodified DNA

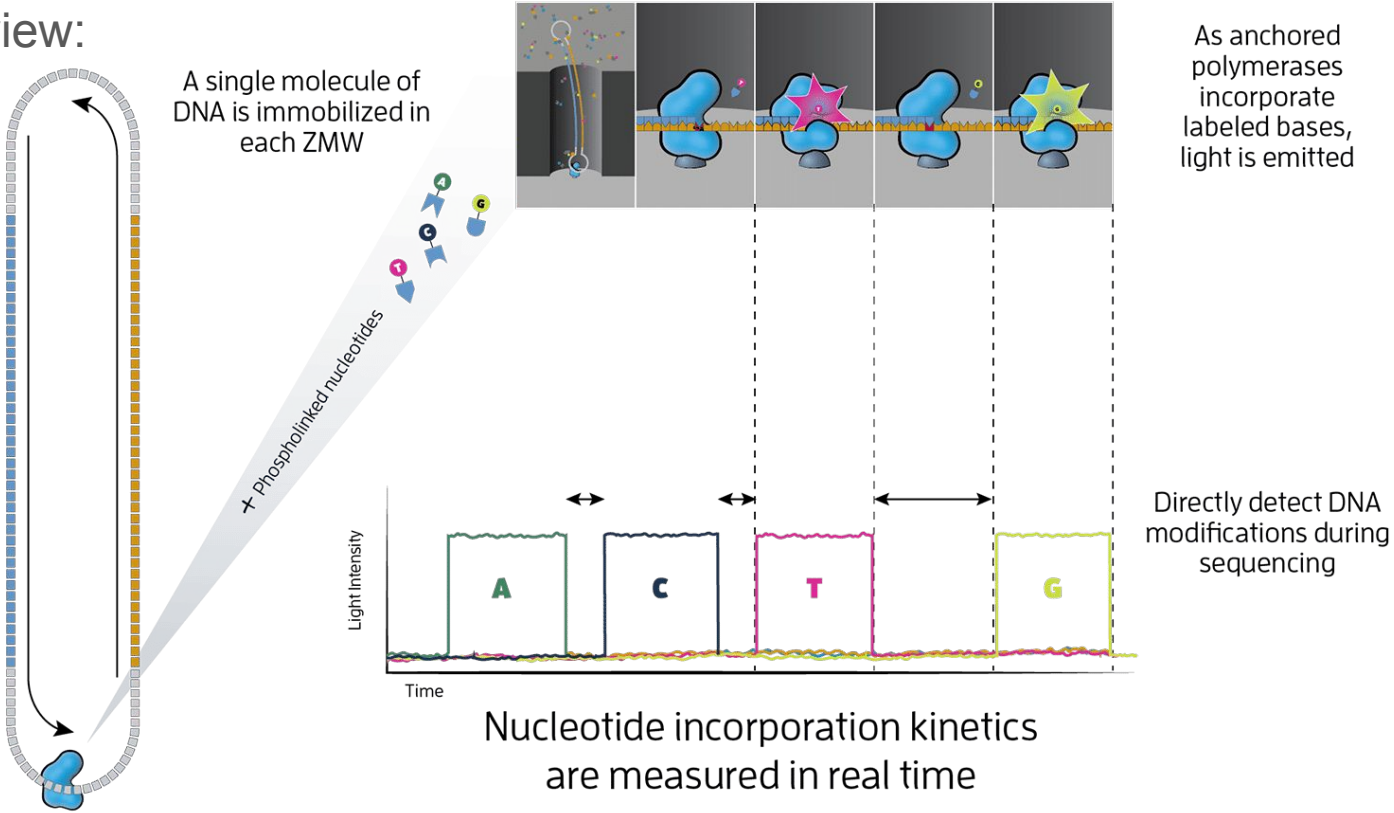




Xu and Seki, Journal of Human Genetics 2019

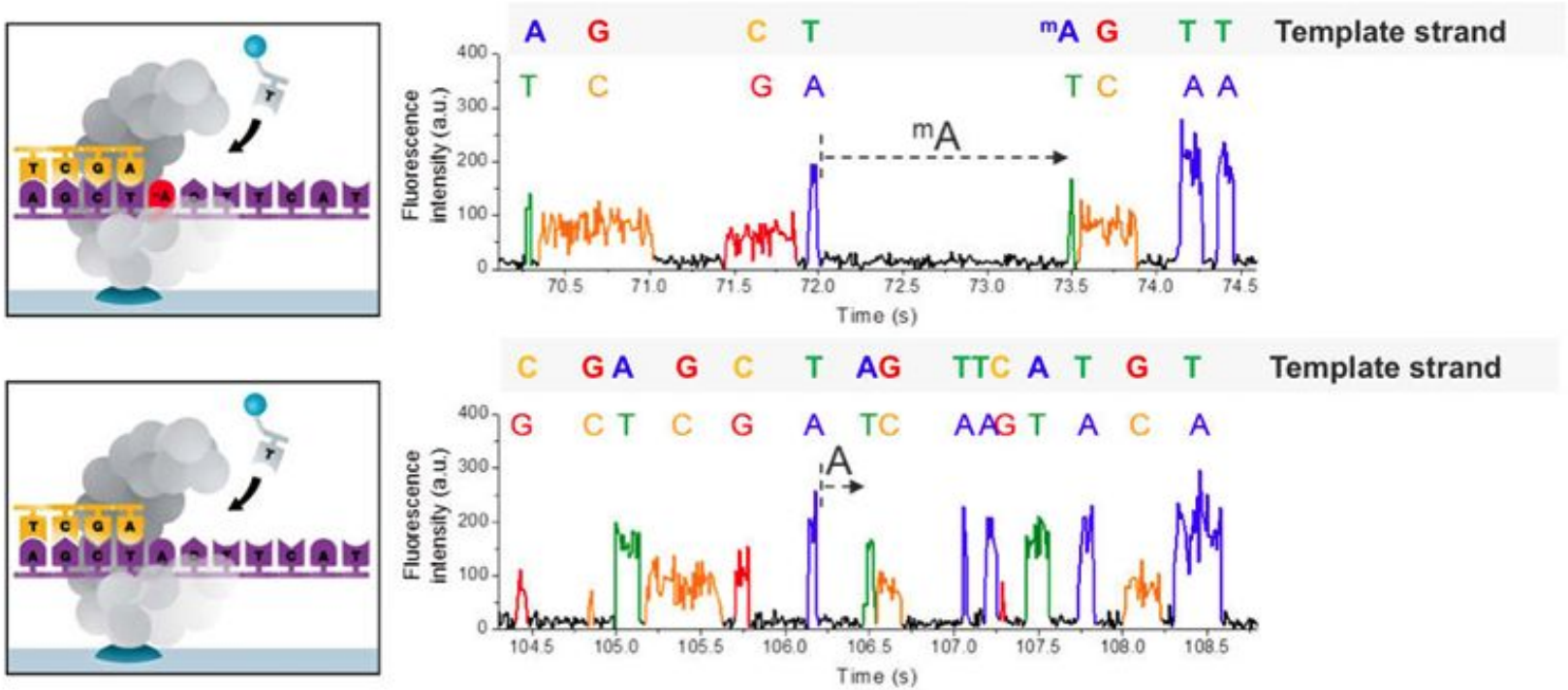
PacBio sequencing can detect DNA methylation natively

Review:

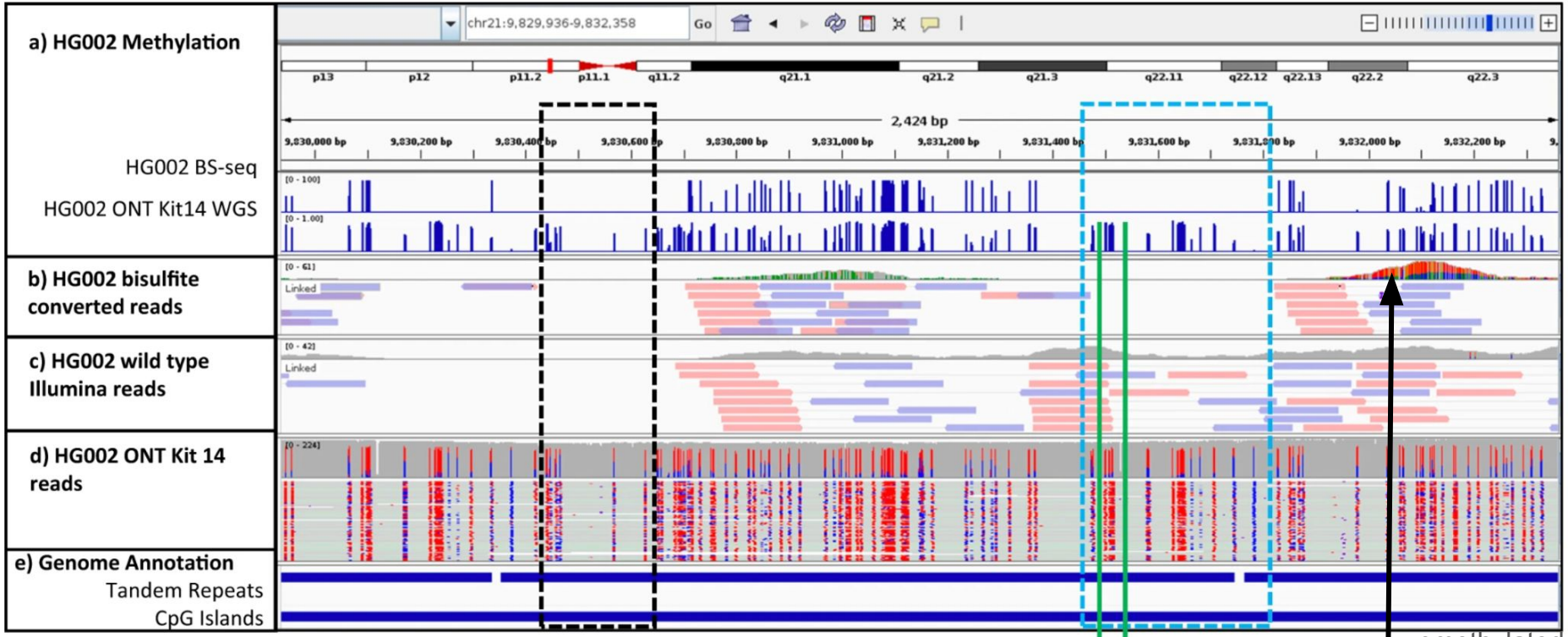


PacBio sequencing can detect DNA methylation natively

Methylated template causes a lag in incorporating a paired nucleotide



Long-read sequencing facilitates methylation detection in complex repeat regions of the genome



Ahsan et al. Nature Communications, 2024

For the next class:

Problem Set 15: Gene regulation and Imprinting

Continue reading in Genetics: From Genes to Genomes

a. Finish Chapter 17.3
b. Start Chapter 22.2

What have chromatin contact maps taught us?

Rao et al. 2014. Cell